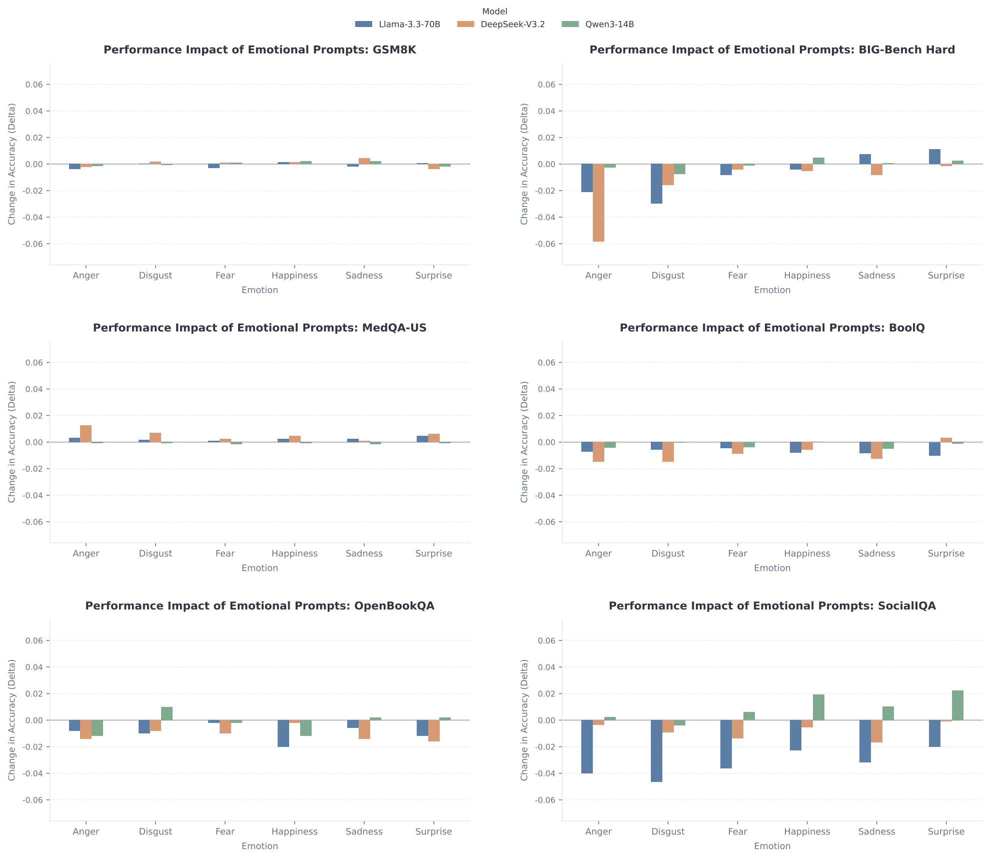

Fig. 3 Effect of static emotional prefixes across benchmark tasks. Accuracy change relative to the matched no-emotion prompt for six prepended emotions across six benchmarks and three backbone models. Each bar isolates the effect of emotional framing while keeping the underlying question unchanged. Most deltas remain close to zero, showing that static emotional prompting usually acts as a mild perturbation rather than a strong performance modifier. The largest dispersion appears in socially grounded settings and a small number of harder reasoning conditions, where the same emotion can help one model and hurt another.

prefixes have limited influence on tightly constrained reasoning and domain-specific multiple-choice prediction. BoolQ and OpenBookQA show somewhat larger but still modest shifts, with OpenBookQA displaying the clearest tendency toward small degradations under fixed emotional prompts. BIG-Bench Hard exhibits occasional drops for particular emotions, indicating that harder reasoning tasks can be somewhat more brittle, although the direction and magnitude of the effect still depend on the model.

The clearest departure from this overall stability appears in SocialIQA. Here, the spread across models and emotions is visibly larger, and the direction of the effect is not consistent. This pattern suggests that emotional context interacts most strongly with tasks that already require reasoning about intentions, beliefs, and interpersonal situations. At the same time, the absence of a consistent best emotion indicates that the effect is heterogeneous rather than universal. Overall, static emotional framing is best characterized as a weak, model- and task-dependent perturbation.

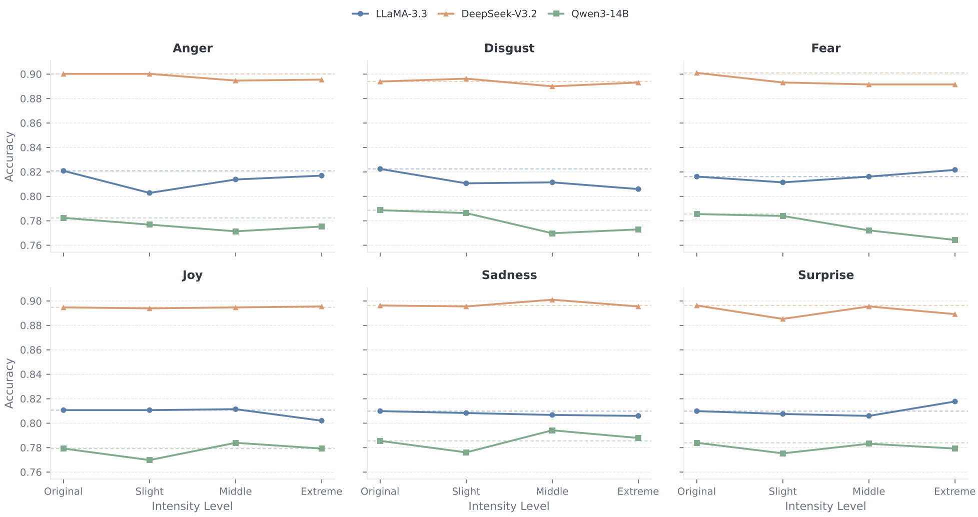

Fig. 4 Effect of emotional intensity on MedQA-US. Accuracy delta relative to the no-emotion baseline as the intensity of a prepended emotion injection statement increases from slight to extreme. All three models remain close to zero across the full range of intensities. Stronger affective wording produces only a mild downward trend and does not induce an abrupt failure regime, indicating that emotional intensity changes the magnitude of the perturbation without qualitatively changing task behavior.

3.2 Stronger emotional wording yields only a small additional shift

We next vary the intensity of emotional wording amplifies its effect on MedQA-US. Figure 4 compares the original no-emotion condition with slight, moderate, and extreme variants across six emotion categories and three backbone models. Overall, the effect of emotional intensity remains limited: accuracy varies only modestly across intensity levels, and stronger wording does not produce a consistent monotonic deterioration in performance. Instead, the pattern is heterogeneous across models and emotions. For some models, emotion pairs exhibit mild decreases as intensity increases, whereas others remain nearly flat or show small rebounds at moderate or extreme levels. This indicates that emotional intensity can modulate model performance, but the induced variation is bounded and directionally inconsistent rather than systematically harmful.

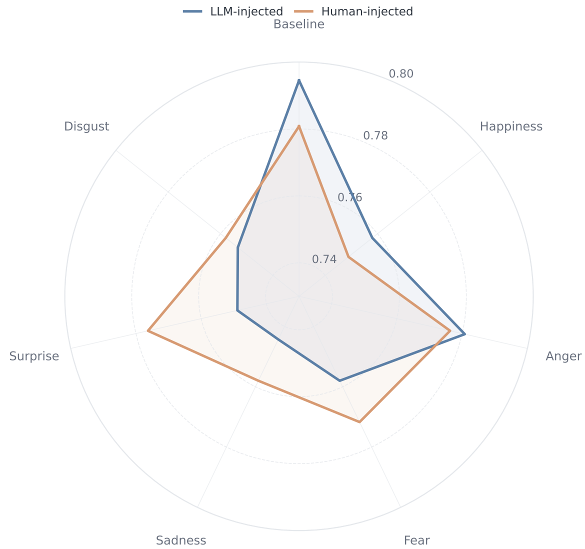

Fig. 5 MedQA-US: human versus LLM emotion injection. Accuracy of Qwen3-14B on the held-out subset under the no-emotion baseline and six emotion conditions, comparing LLM-generated prefixes with human-written prefixes. The two sources produce closely matched accuracies across conditions, and the small differences do not consistently favor one source over the other. The qualitative effect of emotional framing is therefore robust to how the prefix is authored.

3.3 Human- and LLM-authored emotional prefixes lead to the same conclusion

To test whether the benchmark-level conclusions depend on how the emotional text is produced, we compare human-written and LLM-generated prefixes on a held-out MedQA-US subset. Figure 5 shows closely matched accuracies across conditions, with small differences that change sign across emotions. There is no systematic advantage for either source.

This result strengthens the main claim in two ways. Empirically, it suggests that the limited effect of emotional framing is not an artifact of a particular prompt-generation pipeline. Methodologically, it shows that carefully written human stimuli can be used in place of LLM-generated ones without changing the qualitative outcome. This is useful for future studies that require tighter experimental control or clearer interpretation of the emotional manipulation.

3.4 Adaptive emotion selection is more effective than static emotional prompting

The weak average effect of static emotional prompts does not imply that emotional framing contains no usable signal. Figure 6 shows that EmotionRL, which selects an emotion condition on a per-example basis, consistently matches or exceeds the average static-emotion baseline across the five tasks shown. In particular, the learned policy